Instrumental Variables

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Applied Econometrics

- Basic Idea of Instrumental Variable (IV):
  - What if we have a variable that is correlated with X *but not with* Y
  - Then any changes in Y caused by that variable will reflect causal changes by X
- Equation of interest:

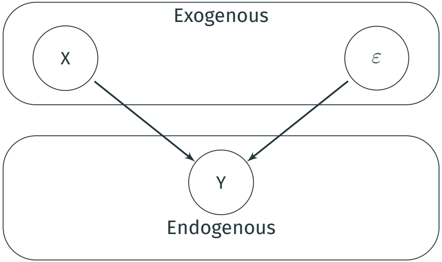
$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

- IVs break X into two pieces that are themselves uncorrelated:

$$X_i = \gamma_0 + \gamma_1 Z_i + \eta_i$$

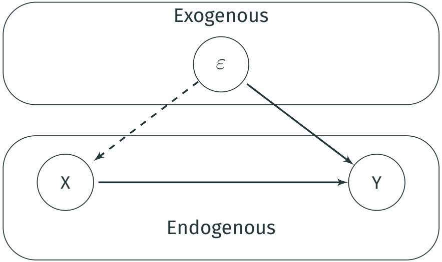
  - A piece that is not correlated with ε ($Cov(Z, \varepsilon) = 0$)
  - A piece that is correlated with ε ($Cov(\eta, \varepsilon) \neq 0$) – source of the endogeneity problem
  - Finally, $Cov(Z, \eta) = 0$
- Z_i is an **instrumental variable**

A “Good” Regression:



- **Exogenous Variables:** Variables in the data that do not cause each other
  - ε is always exogenous, so exogenous also just means variables not correlated with ε
- **Endogenous Variables:** Variables that are determined by exogenous variables in the model
  - ε is always in Y so Y is always endogenous

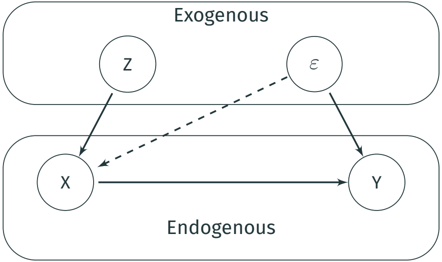
Selection/OVB:



- In this picture X is **endogenous** because ε now causes X as well
- What if there is another exogenous variable that *does not* directly cause Y?

Omitted Variable Bias with Pictures

Instrumental Variable:



- Z causes Y *only indirectly* through X
- We can estimate "causal effect" of Z on Y and this MUST be the causal effect of Z on X and the causal effect of X on Y
  - Mathematically we need to split effect of on Z on Y into effect of Z on X and X on Y
  - Related concept in statistics: Path Analysis

Model:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

- We call a variable Z a a valid **instrumental variable** if the following two conditions hold:

1. **Relevance:** $Cov(X, Z) \neq 0$
   - An arrow from Z to X in the pictures
2. **Exogeneity:** $Cov(\varepsilon, Z) = 0$
   - No arrow from Z to Y or Z to ε in the picture

Key Assumption #1 of Instrumental Variables

- **Relevance:** $Cov(X, Z) \neq 0$
- This assumption just means that X and Z are correlated
- We observe both X and Z, so can easily test this assumption by regressing:

$$X_i = \beta_0 + \beta_1 Z_i + \varepsilon_i$$

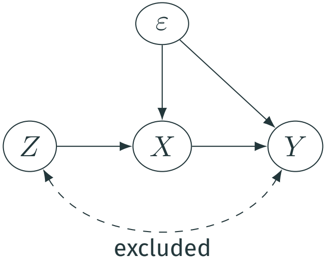
- If $\beta_1 \neq 0$ in regression, we say instrument is relevant

Key Assumptions #2 of Instrumental Variables

- **Exogeneity (or validity):** $Cov(\varepsilon, Z) = 0$
- This assumption means that Z and ε cannot be correlated
- We do not observe ε, so we **cannot** test this assumption
- In general, we need to “defend” this assumption by telling a story about why Z and ε are unlikely to be correlated
- Much like parallel trends or strict exogeneity, these crucial identifying assumptions cannot be tested on their own. However, we can test them against each other. That is, the best we can do is ask the question ”given this exogeneity assumption, does this other exogeneity assumption seem true?” But we can never test an exogeneity assumption in a self-contained way.

IV: Causal Diagram

- Recall the absence of direct path between $Z \to Y$. This is the exclusion restriction.
- $Z \to X$ is the relevance condition.



Idea: There is a path from $Z \to Y$ but only through X (this is the key).

Best Defense of Exogeneity IV Assumption: Randomized Experiment

- Back to the class size example:

$$score_i = \beta_0 + \beta_1 \text{CS}_i + \varepsilon_i$$

- where:
  - $score_i$: Test score of student i
  - CS_i: Class size of student i
- Suppose that we use a coin flip that sends kids that get a "head" to a small class and kids getting a "tails" to big class
  - This is our randomized experiment!
- Let's call the coin flip our instrument Z (where $Z_i = 1$ if heads, $Z_i = 0$ if tails)

Best Defense of Exogeneity IV Assumption: Randomized Experiment

$$score_i = \beta_0 + \beta_1 \text{CS}_i + \varepsilon_i$$

- Is Z (our coin flip) a good instrument?
- **Relevance:** $Cov(CS, Z) \neq 0$? Yes, if $Z_i = 1$ kid gets small class, if $Z_i = 0$ kid gets big class
  - So the regression $CS_i = \beta_0 + \beta_1 Z_i + \varepsilon_i$ will estimate that $\beta_1 < 0$
- **Exogeneity:** $Cov(\varepsilon, Z) \neq 0$? Untestable – so need to tell a story
  - Story: Exogeneity holds because coin flip is random and does not depend on any student or parent characteristics that would affect test scores. Therefore, there is nothing related to Z (besides X) that is also related to test scores, so ε and Z must be uncorrelated.
    This is a good story! It's the "magic" of randomization.

Randomized Experiment as an IV

- So a randomized experiment can be treated as an IV, but there is a big difference in how we evaluate the assumption for an experimental or non-experimental setting
- When running an experiment, the exogeneity assumption is justified if the randomization was implemented in way that was not correlated with anything else that could influence outcomes. Attrition and manipulation can undermine exogeneity in an experimental context if they create a correlation between treatment status and other factors that might influence outcomes. Thus, exogeneity amounts to an assumption about how the randomization was implemented.
- In contrast, the IV exogeneity assumption in a non-experimental context is a broad and vague statement about how the world works; it's much more difficult to interpret and evaluate. The same is true of strict exogeneity or parallel trends.

IV Estimation: Example with a binary $Z \in \{0, 1\}$

Let's begin with the simplest 0/1 model (this time for Z):

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

- X is endogenous but we have a 0/1 instrument, Z
    - Because Z is 0/1 we can focus on group means.
- What assumptions do we need to make?
    - $Cov(X, Z) \neq 0$
    - $Cov(Z, \varepsilon) = 0$
- The intuition from above: solve for the "indirect" effect of Z on Y

The Relationship Between Y on Z

What is the relationship between Y and Z, taking expectations?

$$\begin{aligned}\mathbb{E}(Y|Z) =& E\left(\beta_0 + \beta_1 X + \varepsilon|Z\right) \\ =& \beta_0 + \beta_1 \mathbb{E}(X|Z) + \underbrace{\mathbb{E}(\varepsilon|Z)}_{=0} \\ =& \beta_0 + \beta_1 \mathbb{E}(X|Z)\end{aligned}$$

Since Z is 0/1...

$$\begin{aligned}\mathbb{E}(Y|Z=1) =& \beta_0 + \beta_1 \mathbb{E}(X|Z=1) \\ \mathbb{E}(Y|Z=0) =& \beta_0 + \beta_1 \mathbb{E}(X|Z=0)\end{aligned}$$

Rearranging...

$$\mathbb{E}(Y|Z=1) - \mathbb{E}(Y|Z=0) = \beta_1 \times \left(\mathbb{E}(X|Z=1) - \mathbb{E}(X|Z=0)\right)$$

The IV Regression for a binary $Z \in \{0, 1\}$

From the previous slide we have:

$$\beta_1 = \frac{\mathbb{E}(Y|Z=1) - \mathbb{E}(Y|Z=0)}{\mathbb{E}(X|Z=1) - \mathbb{E}(X|Z=0)}$$

From the LLN we can construct the following estimator:

$$\hat{\beta}_1 = \frac{\bar{Y}_1 - \bar{Y}_0}{\bar{X}_1 - \bar{X}_0}$$

- The subscript refers to $Z = 1$ or $Z = 0$
- We call this the IV estimator

Interpreting the Estimator for binary $Z \in \{0,1\}$

Estimator:

$$\hat{\beta}_1^{IV} = \frac{\bar{Y}_1 - \bar{Y}_0}{\bar{X}_1 - \bar{X}_0}$$

- For a binary X we look at the change in Y over the change in X:

$$\hat{\beta}_1^{OLS} = \frac{\bar{Y}_{X=1} - \bar{Y}_{X=0}}{\bar{X}_{X=1} - \bar{X}_{X=0}} = \Delta\bar{Y}$$

  - For a $0/1$ X, $\Delta\bar{X} = 1$
  - Basically looking at how much Y changes for a change in X
- Intuition for a 0/1 Z
  - The effect of X on Y is still the change in Y over a change in X
  - Use the change in Y *given* Z to shut down the effects of ε
  - Use the change in X *given* Z to get the right scaling

The IV Regression with a Continuous Z

Same model as before:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

- Now assume that Z is continuous
- How to capture the indirect effect of Z?
  - Intuition from OLS: use the covariance between Z and Y as a measure of the relationship:

$$\begin{aligned} Cov(Y, Z) =& Cov(\beta_0 + \beta_1 X + \varepsilon, Z) \\ =& \beta_1 Cov(X, Z) \end{aligned}$$

  - Rearranging:

$$\beta_1 = \frac{Cov(Y, Z)}{Cov(X, Z)}$$

- **Instrumental Variables Estimator:** $\hat{\beta}_1^{IV} = s_{YZ}/s_{XZ}$

Heterogeneous treatment effects I

- Let's consider a simple case with binary treatment $X_i \in \{0, 1\}$ where the treatment effect $\beta_i = \mathbb{E}(Y_{i1} - Y_{i0})$ is heterogeneous.
  - Y_{ix} denotes the potential outcome for i given treatment status $X_i = x$.
- We have a binary instrument $Z_i \in \{0, 1\}$
- Let's also have potential outcomes for treatment status: X_{iz} is the potential treatment status for i when $Z_i = z$.
- Model:

$$Y_i = \beta_0 + \beta_i X_i + \varepsilon_i$$

Heterogeneous treatment effects II

$$Y_i = \beta_0 + \beta_i X_i + \varepsilon_i$$

- Assuming that $\mathbb{E}\left[\varepsilon_i | Z_i = 1\right] - \mathbb{E}\left[\varepsilon_i | Z_i = 0\right] = 0$,

$$\mathbb{E}\left[Y_i | Z_i = 1\right] - \mathbb{E}\left[Y_i | Z_i = 0\right] = \mathbb{E}\left[\beta_i X_i | Z_i = 1\right] - \mathbb{E}\left[\beta_i X_i | Z_i = 0\right]$$

Plugging in our potential outcomes for X,

$$\mathbb{E}\left[Y_i | Z_i = 1\right] - \mathbb{E}\left[Y_i | Z_i = 0\right] = \mathbb{E}\left[\beta_i \left(X_{i1} - X_{i0}\right)\right]$$

$$\mathbb{E}\left[Y_i | Z_i = 1\right] - \mathbb{E}\left[Y_i | Z_i = 0\right] = \mathbb{E}\left[\beta_i \left(X_{i1} - X_{i0}\right)\right]$$

- Given the binary treatment, there are three possible values for $X_{i1} - X_{i0}$:

$$\begin{aligned}
\mathbb{E}\left[\beta_i \left(X_{i1} - X_{i0}\right)\right] &= \text{Pr}\left(X_{i1} - X_{i0} = 1\right) \mathbb{E}\left[\beta_i \left(X_{i1} - X_{i0}\right) | X_{i1} - X_{i0} = 1\right] && \text{(compliers)} \\
&+ \text{Pr}\left(X_{i1} - X_{i0} = 0\right) \mathbb{E}\left[\beta_i \left(X_{i1} - X_{i0}\right) | X_{i1} - X_{i0} = 0\right] && \text{(always/never takers)} \\
&+ \text{Pr}\left(X_{i1} - X_{i0} = -1\right) \mathbb{E}\left[\beta_i \left(X_{i1} - X_{i0}\right) | X_{i1} - X_{i0} = -1\right] && \text{(defiers)}
\end{aligned}$$

- The second term (always/never takers) is necessarily equal to zero. We might also make a **monotonicity assumption** that Pr $\left(X_{i1} - X_{i0} = -1\right) = 0$.
- Ruling out defiers, this could be interpreted to mean that being assigned to treatment can only make you more likely to receive treatment.

Heterogeneous treatment effects IV

- Start with

$$\mathbb{E}\left[Y_i \mid Z_i = 1\right] - \mathbb{E}\left[Y_i \mid Z_i = 0\right] = \mathbb{E}\left[\beta_i \left(X_{i1} - X_{i0}\right)\right]$$

- Monotonicity means:

$$\mathbb{E}\left[\beta_i \left(X_{i1} - X_{i0}\right)\right] = \text{Pr}\left(X_{i1} - X_{i0} = 1\right) \mathbb{E}\left[\beta_i \left(X_{i1} - X_{i0}\right) \mid X_{i1} - X_{i0} = 1\right]$$

- Also note that $Pr\left(X_{i1} - X_{i0} = 1\right) = E\left[X_i \mid Z_i = 1\right] - E\left[X_i \mid Z_i = 0\right]$
- Combining the above,

$$\begin{aligned}\frac{\mathbb{E}\left[Y_i \mid Z_i = 1\right] - \mathbb{E}\left[Y_i \mid Z_i = 0\right]}{\mathbb{E}\left[X_i \mid Z_i = 1\right] - \mathbb{E}\left[X_i \mid Z_i = 0\right]} &= \mathbb{E}\left[\beta_i \left(X_{i1} - X_{i0}\right) \mid X_{i1} - X_{i0} = 1\right] \\ &= \mathbb{E}\left[\beta_i \mid X_{i1} - X_{i0} = 1\right]\end{aligned}$$

- We now have that our IV estimator equals the **local average treatment effect**, which means the average effect on the individuals whose treatment status is changed by the instrument.

IV Example

- Suppose that we are interested in investigating the effect of studying on grades:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

- where
  - Y_i: is GPA
  - X_i: is study time (hours per day)

- We expect that our OLS estimator $\widehat{\beta}_1$ will be severely biased here (why?)

- To get rid of OVB, we shall use an IV
- One possible IV: whether your roommate has a N64 (or playstation/whatever video game console kids use today)
- Important feature: Roommates are randomly assigned in college
    - At least at Berea College (Kentucky) where this example comes from (see Stinebrickner and Stinebrickner (2008))

Instrument Validity

- First thing for any instrument is to think of validity. Two conditions:
    - Instrument is $N64$, which is indicator variable that your roommate has N64

1. Relevance: $Cov(Z, X) \neq 0$; here $Cov(N64, study) \neq 0$
    - Seems likely to hold as everyone prefers playing Mario Kart to studying
    - Testable

2. Exogeneity: $Cov(Z, \varepsilon) = 0$; here $Cov(N64, \varepsilon) = 0$
    - Untestable
    - Is it plausible?

Thinking About Exogeneity

- Exogeneity: $Cov(Z, \varepsilon) = 0$
- "Storytime" should talk about how two things hold
  1. People select into Z in some manner that is likely to be correlated with Y
     - Concern: People that pick roommates that are "fun" and have N64s are likely people who do not care too much about grades
  2. Z only affects Y through X
     - Concern: Having a roommate with a video game affects your GPA through other means than affecting your study hours
- Either story "invalidates" the instrument (but in different ways)

Thinking About Exogeneity

1. Selection story: there is an omitted variable related to both Z and Y
   - Example: omitted variable is effort because students that really put in a lot of effort make sure they do not get a roommate with N64

2. Other channel story:
   - Possible Story 1: Mario Kart helps me grasp physics, so I ace my physics exam
     - so Z directly affects Y *independent* of X
   - Possible Story 2: Other people hang out in our room due to our N64, making it really loud and so I cannot study effectively
     - so Z affects Y through *another* X

- Conceptually, there are two aspects to the exogeneity assumption: **randomization** and **exclusion**. Be careful: even when an IV is clearly random, it might not be excludable.

IV Math

IV Directly into Model

- Suppose that our IV assumptions hold
- Then we can just directly replace X with our IV in our model:

$$Y_i = \pi_0 + \pi_1 Z_i + \varepsilon_i$$

- where
  - Y_i: is GPA
  - Z_i: is whether roommate has N64
- **Interpretation of $\hat{\beta}_1$:** Having a roommate with a N64 **causes** a $\hat{\beta}_1$ decrease in GPA
  - Causal effect because validity of instrument, $cov(Z, \varepsilon) = 0$, implies OLS is consistent for above equation. Of course we could question the instrument's validity.
- But we want the effect of *studying* on GPA!

A popular IV estimator is Two Stage Least Squares (2SLS)

- This effectively runs regression $Y_i = \beta_0 + \beta_1 Z_i + \varepsilon_i$, but *scales* β_1 by how much Z affects X
  - If Z affects X at one-to-one rate, β_1 in above regression will capture effect of X
  - If Z affects X only a little, we need to scale β_1 up a lot to say what effect of X on Y is
- We therefore estimate in two steps:
  1. Regress X on Z to capture effect of Z on X
  2. Regress fitted values of step 1 to capture effect of X on Y
- Under IV assumptions, this gives us **causal** effect of X on Y

- Step 1: regress

$$X_i = \pi_0 + \pi_1 Z_i + \eta_i$$

- Step 2: take your predicted values from Step 1, $\hat{X}_i$, and regress

$$Y_i = \gamma_0 + \gamma_1 \hat{X}_i + \varepsilon_i$$

- Under IV assumptions, step 1 finds the “good part” of X (that is not related to ε), and then step 2 takes that “good part” of X as a regression to find the causal relationship between X and Y

- Why is $\hat{\beta}_1^{2SLS} = \frac{Cov(Y,Z)}{Cov(X,Z)}$?
- True model:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

- Take covariance of both sides with respect to Z:

$$Cov(Y_i, Z_i) = Cov(\beta_0 + \beta_1 X_i + \varepsilon_i, Z_i)$$
$$Cov(Y_i, Z_i) = Cov(\beta_0, Z_i) + Cov(\beta_1 X_i, Z_i) + Cov(\varepsilon_i, Z_i)$$
$$Cov(Y_i, Z_i) = 0 + \beta_1 Cov(X_i, Z_i) + 0 \ \textit{ since } Cov(\varepsilon_i, Z_i) \textit{ by assumption}$$
$$\implies \beta_1 = \frac{Cov(Y, Z)}{Cov(X, Z)}$$

Single Variable IV in R: Manually

- Data:
  - Y_i: is GPA
  - X_i: is study time (hours per day)
  - Z_i: is indicator for roomate having N64
- Running 2SLS by doing both steps manually:
- Step 1: Regress $X_i = \pi_0 + \pi_1 Z_i + v_i$ and get predicted study time, $\hat{X}_i$
  - `Data$xHat = predict(feols(X~Z, data=Data))`
- Step 2: Regress $y_i = \gamma_0 + \gamma_1 \hat{X}_i + \varepsilon_i$
  - `m3 = feols(Y~xHat, data=Data)`
  - `summary(m3, vcov= "HC1"))`
    - Note: Standard errors will be wrong!
  - `predict` gives the fitted values of X from the first stage
  - Then `feols` can be used (again) to get the second stage
  - Problem: the standard errors (even robust) will be wrong

Doing Single Variable IV in R: All at once

R can run 2SLS all at once:

- `m4 = feols(Y ~ W | X ~ Z, data=Data)`
- First formula: Regress Y on exogenous controls W (and endogenous X)
- Second formula: Write first-stage regression for endogenous X (included in second stage automatically)
- `feols` will also adjust standard errors

This is basically why we use `feols` whenever we can.

- Consider a multiple regression equation

$$y = \beta X + \varepsilon$$

  The IV assumptions are now that
  - $Cov\left(z_i, \varepsilon_i\right) = 0$
  - $Cov\left(z_i, x_i\right)$ has *full rank*, i.e., rank K, where K is the number of regressors in **x**
- We can have multiple regressors as well as instruments. Some regressors can also be instruments; if $z_i = x_i$ then we can just use OLS.
- Regressors that are instruments are **exogenous regressors**. Regressors that are not instruments are **endogenous regressors**.
- For the rank assumption, we need at least one instrument for each endogenous regressor.

2SLS with Matrix Notation

- The 2SLS formula is now

$$b_{2SLS} = \left(\hat{X}'\hat{X}\right)^{-1}\hat{X}'y$$

$$\text{with } \hat{X} = Z\left(Z'Z\right)^{-1}Z'X$$

- With homoscedasticity, the asymptotic variance of the estimator is

$$Var\left(b_{2SLS}\right) = \frac{s^2}{n}\left[Z'X\left(X'X\right)^{-1}X'Z\right]^{-1}$$

- Note: this (asymptotically correct) estimate of the variance is not equal to the (incorrect) formula we would get from naïvely applying OLS formulas to the second-stage regression.

IV w/ controls

- `m5 = feols(workedm ~ morekids + boy1st + boy2nd + black + hispan + othrace + agem1 + agefstm | samesex ~ boy1st + boy2nd, data=babyData)`
- Second stage | First stage
- Don't include endogenous X before the pipe (only exogenous controls)
- Need to include excluded instruments in the second equation!
- Syntax differs for different packages $\rightarrow$ Be careful!

Comparing 2SLS Variance to OLS Variance

For the single-variable case, we can show that

$$Var(\hat{\beta}_{2SLS}) = \frac{1}{N}\frac{\sigma_\varepsilon^2}{\sigma_X^2 R_{XZ}^2}\mathbf{X}'\mathbf{e}_{OLS} = \mathbf{0}$$

OLS variance under homoscedasticity: $Var(\hat{\beta}^{OLS}) = \frac{1}{N}\frac{\sigma_\varepsilon^2}{\sigma_X^2}$

- Since $R_{XZ}^2 \leq 1$, IV variance is larger
- The higher first stage R_{XZ}^2, the lower the variance. If X and Z are collinear, variance is equivalent to OLS.
- If $R_{XZ}^2 \approx 0$ then variance blows up!

Unbiasedness “Proof”

$$
\begin{aligned}
\mathbb{E}[\hat{\beta}_1^{2SLS}] &= \mathbb{E}\left[\frac{\sum_{i=1}^{N}(z_i - \bar{z})(y_i - \bar{y})}{\sum_{i=1}^{N}(x_i - \bar{x})(z_i - \bar{z})}\right] \\
&= \beta_1 + \mathbb{E}\left[\frac{\sum_{i=1}^{N}(z_i - \bar{z})\varepsilon_i}{\sum_{i=1}^{N}(x_i - \bar{x})(z_i - \bar{z})}\right] = \beta_1 + 0
\end{aligned}
$$

Checking The IV Assumptions I

- The two major assumptions:
    1. Relevance: $Cov(X,Z) \neq 0$, or that $\mathbb{E}\left[\mathbf{z}_i \mathbf{x}_i'\right]$ has full rank
    2. Exogeneity: All instruments must be uncorrelated with ε
- First, let's talk about the relevance assumption.
- It's easy to check that the sample analog to $Cov(X,Z) \neq 0$, or the analogous matrix condition.
- The more practically relevant concern is that the covariance will be *close* to zero. This is called **weak instruments**.
- How do we test and deal with weak instruments in practice?

- In finite samples there's a possibility that the sample covariance

$$\sum_{i=1}^{N}(x_i - \bar{x})(z_i - \bar{z})$$

  is close to zero, especially if the true covariance is relatively small.

- Dividing by zero is *bad*.

- Note we don't have the same problem in OLS as long as we have variation in the regressors.

- There is **big** problem with IV estimation: 2SLS performs **terribly** in small samples
- Intuition: Instrumental variables relies on $Corr(X, Z) \neq 0$
  - If $Cov(X, Z) = 0$ then IV estimator is infinite!
- In practice, very rarely will $Cov(X, Z) = 0$ in data (just randomness)
  - Hence, IV will usually “work” with any choice of X and Z
  - Realistically, sometimes $Cov(X, Z)$ is small if Z only explains a small portion of X
- If $Cov(X, Z)$ is small we say that Z is a weak instrument

Weak Instruments and Small Sample Bias

Suppose that Z is *a bad instrument* in that $Cov(Z, \varepsilon)$ are correlated

- Similar to OVB

$$\hat{\beta}_1^{IV} \to \beta_1 + \frac{Cov(Z, \varepsilon)}{Cov(Z, X)} \times \frac{\sigma_\varepsilon}{\sigma_X}$$

- Compare to OLS case (biased because of OVB):

$$\hat{\beta}_1^{OLS} \to \beta_1 + Cov(X, \varepsilon) \times \frac{\sigma_\varepsilon}{\sigma_X}$$

- Which estimator is better here?
  - IV's advantage is $Corr(Z, \varepsilon)$ should be smaller than $Corr(X, \varepsilon)$
  - But if instrument is weak $Corr(Z, X) \approx 0$, then bias blows up even if $Corr(Z, \varepsilon)$ small
- Danger of weak instruments:
  - Very small bias in Z can be greatly magnified if instruments are weak
  - Amplified bias may have different sign than OLS

To do inference we rely on the CLT:

$$\hat{\beta}^{IV} \sim \mathcal{N}\left(\beta, Var\left(\hat{\beta}^{IV}\right)\right)$$

- First issue: the Normal Approximation fails unless N is huge! (the CLT does not work well)
  - Intuition: $\hat{\beta}^{IV} = Cov(Y, Z)/Cov(X, Z)$.
  - If $Cov(X, Z) \approx 0$ then small changes in estimated covariance can have huge effects on estimated coefficients

- Second issue: even if the normal approximation works the variance explodes!
  - Intuition from homoscedastic case: $Var(\hat{\beta}^{IV})) = \sigma^2/(N\sigma_X^2 R_{XZ}^2)$
  - If the R^2 of X on Z is very small, then variance is very large

How To Deal With Weak Instruments?

What are we to do if instruments are weak?

- Easiest: get more instruments or better instruments
- Harder: adjust how to do inference
    - Turns out that if instruments are weak, asymptotic distribution of $\hat{\beta}$ will be the *ratio* of two *correlated* normal random variables
    - We will not pursue this (uses LIML)
- The good news: We can test $Cov(X, Z)$ by regressing $x_i = \pi_0 + \pi_1 Z_i + \eta_i$
    - Since Z and X are data, we can just see if $\hat{\pi}_1$ is far from zero
- **Rule of Thumb:** Only use instrument if t-stat of null hypothesis that $\pi_1 = 0$ is ≥ 10 (much bigger than stat significance of 1.96!)

Some clarification:

- With only one endogenous variable, the $F > 10$ cutoff refers to the first-stage regression of that endogenous variable on all the exogenous variables. The null hypothesis for that F-statistic is that the coefficients on all the excluded instrumental variables are zero.
- With multiple endogenous variables, there are multiple first-stage regressions, and we should be wary if *any* of them suggests a weak instrument problem.
- Many have argued that the $F > 10$ cutoff is not stringent enough. See, for example, https://arxiv.org/abs/2010.05058Lee, McCrary, Moreira, and Porter

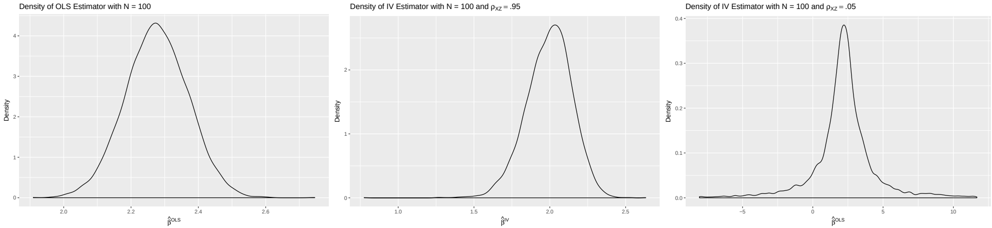
The model:

$$
\begin{aligned}
Y_i =& \beta_0 + \beta_1 X_i + \varepsilon_i \\
X_i =& \pi_0 + \pi_1 Z_i^{(1)} + \pi_2 Z_i^{(2)} + \ldots + \pi_m Z_i^{(m)} + \eta_i
\end{aligned}
$$

- $\pi \approx 0 \Rightarrow$ because of *sampling error*, there is a good chance that $\hat{\pi} < 0$ even if $\pi > 0$
- If $\hat{X}$ is the wrong *sign* or close to *zero* then it is very likely that $\hat{\beta}^{IV}$ will behave poorly
- **NB:** As $N \to \infty$, $\hat{\pi} \to \pi$, so weak instruments are a *small sample problem*.

$$Y = 1 + 2X + \varepsilon$$

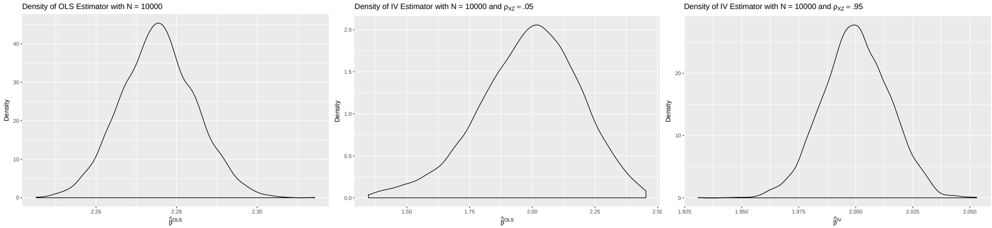
Figure 1: Distribution of $\widehat{\beta}$ for OLS, strong IV, weak IV with $N = 100$



- OLS is biased with $\mu = 2.27$ and $\sigma^2 = .008$
- Strong IV is unbiased (asymptotically) with $\mu = 1.99$ and $\sigma^2 = .024$
- Weak IV is unbiased (asymptotically) with $\mu = 2.05$ but $\boldsymbol{\sigma^2 = 1653.11}$

$$Y = 1 + 2X + \varepsilon$$

Figure 2: Distribution of $\widehat{\beta}$ for OLS, weak IV, strong IV with $N = 10000$



- OLS remains biased with $\mu = 2.28$ and $\sigma^2 = .000085$
- Strong IV is consistent with $\mu = 1.99$ and $\sigma^2 = .00021$
- Weak IV is consistent with $\mu = 2.05$ and even with $N = 10k$, $\sigma^2 = .05$

- With weak IV, CLT fails and behaves poorly even if N is large
- Remember that for testing we want to reject the null hypothesis incorrectly 5% of the time
  - When the CLT fails, our trusty 1.96 critical value is way off base
  - Example rejection probabilities for test that $\hat{\beta} = 2$

Hypothetical Situation	$P(\|t\| > 1.96)$
Weak IV, $N = 100$	0.3%
Strong IV, $N = 100$	4.2%
Weak IV, $N = 10000$	3.2%
Strong IV, $N = 10000$	5.1%

  - Goal should be 5% but for weak IV with N small, almost never reject; still only 3% even for N large—this is a bad test!

Checking The IV Assumptions II

- The two major assumptions:
  1. Relevance: $Cov(X, Z) \neq 0$, or that $\mathbb{E}\left[\mathbf{z}_i \mathbf{x}_i'\right]$ has full rank
  2. Exogeneity: All instruments must be uncorrelated with ε
- Now, let's talk about the exogeneity assumption.
- With one IV, it's impossible to test exogeneity, hence the need to evaluate exogeneity by thinking about whether some unobserved variables might be correlated with the instrument.

- When the number of regressors equals the number of instruments, the model is **just identified**. When there are more instruments than regressors, the model is **overidentified**.
- For a just identified model, the 2SLS estimator simplifies to

$$\mathbf{b}_{IV} = \left(\mathbf{Z}'\mathbf{X}\right)^{-1}\mathbf{Z}'\mathbf{y},$$

which is often called simply the **IV estimator**.

A Note on Moments I

- Recall the property of the OLS estimator:

$$\mathbf{X}'\mathbf{e}_{OLS} = \mathbf{0}$$

where

$$\mathbf{e}_{OLS} = (\mathbf{y} - \mathbf{X}\mathbf{b}_{OLS})$$

- Similarly, for the IV estimator,

$$\mathbf{Z}'\mathbf{e}_{IV} = \mathbf{0}$$

where

$$\mathbf{e}_{IV} = (\mathbf{y} - \mathbf{X}\mathbf{b}_{IV})$$

- (Derivation on board)

$$\mathbf{X}'\mathbf{e}_{OLS} = \mathbf{0}$$
$$\mathbf{Z}'\mathbf{e}_{IV} = \mathbf{0}$$

- A consequence of this is that we can't test the assumption that the instrument(s) and error terms are uncorrelated by looking at the correlation between the instrument(s) and residuals.
- Similarly, the residuals from OLS don't allow us to test whether the error terms are correlated with the regressors.

Tests of Exogeneity?

One possibility is the Durbin-Wu-Hausman Test which compares $\Delta = \beta_{OLS} - \beta_{2SLS}$ under the null that X is exogenous (Z is assumed exogenous).

$$
\begin{aligned}
H_0 :& \beta_{OLS}, \beta_{2SLS} \text{ are consistent,} \quad \beta_{OLS} \text{ is efficient} \\
H_1 :& \beta_{2SLS} \text{ is consistent, } \beta_{OLS} \text{ is not .}
\end{aligned}
$$

This gives a chi-squared test statistic:

$$T = (\hat{\beta}_{2SLS} - \hat{\beta}_{OLS})^T \left(V(\hat{\beta}_{2SLS}) - V(\hat{\beta}_{OLS})\right)^\dagger (\hat{\beta}_{2SLS} - \hat{\beta}_{OLS}) \sim \chi^2_K$$

This uses the pseudo inverse of the variance matrix where K is the rank of that matrix.

Idea: Reject H_0 if $\|\Delta\| \gg 0$ (2SLS and OLS disagree).

Example in R: Checking Angrist & Evans Validity

```r
 # First Stage Test in Angrist & Evans
flm1 = feols(morekids ~ samesex + black + hispan, data=babyData)
lh1 = linearHypothesis(flm1,"samesex = 0")

# First Stage Test in AE w/ Multiple Instruments
flm2 = lm(morekids ~ boyBoy + girlGirl + black + hispan, data=babyData)
lh2 = linearHypothesis(flm2,c("boyBoy","girlGirl"))
```

Results:

- With “Same Sex”: $F = 25.874$
- With “BoyBoy” and “GirlGirl”: $F = 12.94$

Checking Instrument Validity II

Exogeneity Assumption:

$$Corr(Z_i, \varepsilon_i) \quad \text{for all } Z$$

- If exogeneity violated then $\hat{X}$ from the first stage will still be correlated with $\varepsilon \Rightarrow$ omitted variables bias
- Usual case: need to *assume* exogeneity
- With multiple instruments can [with some limitations] test exogeneity
  - Intuition: if $Z^{(1)}$ and $Z^{(2)}$ are both valid instruments then using them separately should produce the same value of $\hat{\beta}$
  - If $\hat{\beta}$ different for different Z's then one Z must be bad

Operationalizing the Intuition

Procedure to Test Overidentifying Restrictions

1. Estimate the model using 2SLS
2. Compute e_i with X_i (not $\hat{X}_i$):

$$e_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i^{(1)} - ...\hat{\beta}_m X_i^{(k)}$$

3. Regress e_i on all Z's and exogenous variables:

$$e_i = \alpha_0 + \alpha_1 Z_i^{(1)} + ... + \alpha_m Z_i^{(m)} + Exogenous + \varepsilon_i$$

4. Compute the F statistic for $\alpha_1 = ... = \alpha_m = 0$ but do *not* do the standard F test (degrees of freedom would be wrong)
5. Compute $J = mF$, where m is the number of instruments

Operationalizing the Intuition

- Under the null hypothesis, $J \sim \chi^2_{m-k}$, where k is the number of regressors.
  - Notice: We use $m - k$ degrees of freedom, not m
  - Why? Because e is a function of $\hat{\beta}$, which has k parameters
  - If at least one $\alpha \neq 0$ then at least one instrument is endogenous
- The J-test:
  - Calculate the J statistic
  - Compare the J statistic to the critical value for a χ^2_{m-k} distribution
  - Instruments are not exogenous if the test is rejected (so we want the null to not be rejected)
- This is know as Sargan's J test, and it can be implemented using the AER package.
- We can only do this if over-identified
  - If exactly identified, then $Cov(e, Z) = 0$ as we argued above

Interpreting J Test Results

- What happens if we reject the null?
  - The α that is non-zero does not necessarily indicate the problematic instrument – e is itself not unbiased if IV assumptions violated
  - Need to think hard about which instruments may be invalid
- What happens if we do not reject the null?
  - None of your instruments contradict each other by predicting very different e's
  - Does NOT imply that instruments must be valid—they could all be invalid

New Model:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$
$$X_i = \alpha_0 + \alpha_1 Y_i + \nu_i$$

- In many economic environments, variables both determine each other "in equilibrium" or are determined together "jointly"
- A model where X and Y "cause" each other is called a **simultaneous equations model**
- Examples:
  - Angrist & Evans is an example of this: labor supply and fertility decisions are determined jointly, there is no one-way channel
  - Supply and demand is the canonical example: supply equations and demand equations are of interest but in equilibrium only observe $Q_S = Q_D$
  - Also common in macroeconomic contexts: interest rates, unemployment rates, etc.

Direct effect of X on Y:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

First let's solve for X and Y *jointly*

$$\begin{aligned} X_i &= \alpha_0 + \alpha_1 Y_i + \nu_i \\ &= \alpha_0 + \alpha_1(\beta_0 + \beta_1 X_i + \varepsilon_i) + \nu_i \\ &= \alpha_0 + \beta_0\alpha_1 + \alpha_1\beta_1 X_i + \alpha_1\varepsilon_i + \nu_i \end{aligned}$$

Which implies...

$$X_i = \frac{\alpha_0 + \alpha_1\beta_0 + \nu_i}{1 - \alpha_1\beta_1} + \underbrace{\frac{\alpha_1}{1 - \alpha_1\beta_1} \times \varepsilon_i}_{OVB}$$

Direct effect of X on Y:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

What happens with OLS? Think about the OVB equation:

$$\begin{aligned}
\hat{\beta}_1^{OLS} &\rightarrow \beta_1 + \frac{Cov(X, \varepsilon)}{Var(X)} \\
&= \beta_1 + \frac{Cov\left(\frac{\alpha_0 + \alpha_1 \beta_0 + \nu_i}{1 - \alpha_1 \beta_1} + \frac{\alpha_1}{1 - \alpha_1 \beta_1} \times \varepsilon_i, \varepsilon_i\right)}{Var\left(\frac{\alpha_0 + \alpha_1 \beta_0 + \nu_i}{1 - \alpha_1 \beta_1} + \frac{\alpha_1}{1 - \alpha_1 \beta_1} \times \varepsilon_i\right)} \\
&= \beta_1 + \frac{\frac{\sigma_{\varepsilon\nu} + \alpha_1 \sigma_\varepsilon^2}{1 - \alpha_1 \beta_1}}{\frac{\sigma_\nu^2 + \alpha_1^2 \sigma_\varepsilon^2 + 2\alpha_1 \sigma_{\varepsilon\nu}}{(1 - \alpha_1 \beta_1)^2}} \\
&= \beta_1 + (1 - \alpha_1 \beta_1) \times \frac{\alpha_1 \sigma_\varepsilon^2 + \sigma_{\varepsilon\nu}}{\alpha_1^2 \sigma_\varepsilon^2 + \sigma_\nu^2 + 2\alpha_1 \sigma_{\varepsilon\nu}}
\end{aligned}$$

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&= \beta_1 + \frac{\frac{\sigma_{\varepsilon\nu}+\alpha_1\sigma_\varepsilon^2}{1-\alpha_1\beta_1}}{\frac{\sigma_\nu^2+\alpha_1^2\sigma_\varepsilon^2+2\alpha_1\sigma_{\varepsilon\nu}}{(1-\alpha_1\beta_1)^2}} \\
&= \beta_1 + (1-\alpha_1\beta_1) \times \frac{\alpha_1\sigma_\varepsilon^2 + \sigma_{\varepsilon\nu}}{\alpha_1^2\sigma_\varepsilon^2 + \sigma_\nu^2 + 2\alpha_1\sigma_{\varepsilon\nu}}
\end{aligned}$$

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&= \beta_1 + \frac{\frac{\sigma_{\varepsilon\nu}+\alpha_1\sigma_\varepsilon^2}{1-\alpha_1\beta_1}}{\frac{\sigma_\nu^2+\alpha_1^2\sigma_\varepsilon^2+2\alpha_1\sigma_{\varepsilon\nu}}{(1-\alpha_1\beta_1)^2}} \\
&= \beta_1 + (1-\alpha_1\beta_1) \times \frac{\alpha_1\sigma_\varepsilon^2 + \sigma_{\varepsilon\nu}}{\alpha_1^2\sigma_\varepsilon^2 + \sigma_\nu^2 + 2\alpha_1\sigma_{\varepsilon\nu}}
\end{aligned}$$